Cartesian Products

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Exercises

1. Show there is a bijective correspondence of $A \times B$ and $B \times A$.

Proof. It is claimed that the function $f : A \times B \to B \times A$ defined as $f((a, b)) = (b, a)$. To show f is surjective, consider $(b, a) \in B \times A$. Then, by definition $b \in B$ and $a \in A$. As such, $(a, b) \in A \times B$, and therefore $f((a, b)) = (b, a)$. Thus, since (b, a) was an arbitrary element of $B \times A$ it has been shown that for any element in $s \in B \times A$ there exists at least one element $t \in A \times B$ such that $f(t) = s$. Hence, f is surjective.

Consider now $(a, b), (c, d) \in A \times B$ such that $f((a, b)) = f((c, d)) = (s, t)$. By definition of f, note $a = t$ and $c = t$ implying $a = c$. Similarly, it must be that $b = d$, and thus $(a, b) = (c, d)$. Therefore, f is injective and it can be thus concluded that f is bijective. □

2.

(a) Show that if $n > 1$ there is a bijective correspondence of

$$A_1 \times \cdots \times A_n \quad \text{with} \quad (A_1 \times \cdots \times A_{n-1}) \times A_n.$$

Proof. If $n = 2$, note $A_1 \times A_2 = (A_1) \times A_2$ and thus there exists a bijective correspondence between both sets, in particular this bijective map is the identity map. Thus, assume $n > 2$. We claim that $f : A_1 \times \cdots \times A_n \to (A_1 \times \cdots \times A_{n-1}) \times A_n$ defined by $(a_1, \ldots, a_n) \mapsto ((a_1, \ldots, a_{n-1}), a_n)$ is bijective. We begin by showing f is injective. Let $(a_1, \ldots, a_n)$ and $(b_1, \ldots, b_n)$ be elements in the deomain set such that they both map to $((c_1, \ldots, c_{n-1}), c_n)$. By construction, $a_n = c_n$ and $b_n = c_n$ implying $a_n = b_n$. Similarly, $a_1, \ldots, a_{n-1}$ must all equal the corresponding element in $(c_1, \ldots, c_{n-1})$, i.e., $a_1 = c_1$ and so on. In the same way, $b_i = c_i$ for $1 \leq i \leq n - 1$. Therefore, $a_i = b_i$ for $1 \leq i \leq n-1$, and we thus conclude $(a_1, \ldots, a_n) = (b_1, \ldots, b_n)$ making f injective.
To show f is surjective, consider the element $((c_1, \ldots, c_{n-1}), c_n)$ in the set being mapped onto by f. By construction, $c_i \in A_i$ for $1 \leq i \leq n$. Thus, $(c_1, \ldots, c_n) \in A_1 \times \cdots \times A_n$. By definition of f, $f(c_1, \ldots, c_n) = ((c_1, \ldots, c_{n-1}), c_n)$. Thus, we have shown f is also surjective, and therefore we conclude f is bijective. □

(b) Given the indexed family $\{A_1, A_2, \ldots\}$ let $B_i = A_{2i-1} \times A_{2i}$ for each positive integer i. Show there is a bijective correspondence of $A_1 \times A_2 \times \cdots$ with $B_1 \times B_2 \times \cdots$.
Proof. Note $B_1 \times B_2 \times \cdots$ is equivelent to the set $(A_1 \times A_2) \times (A_3 \times A_4) \times \cdots$ by construction. □

3. Let $A = A_1 \times A_2 \times \cdots$ and $B = B_1 \times B_2 \times \cdots$.

(a) Show that if $B_i \subset A_i$ for all i, then $B \subset A$. (Strictly speaking, if we are given a function mapping the index set $\mathbb{Z}_+$ into the union of the sets B_i, we must change its range before it can be considered as a function mapping $\mathbb{Z}_+$ into the union of the sets A_i. We shall ignore this technicality when dealing with Cartesian products).
(b) Show the converse of (a) holds if B is non-empty.
(c) Show that if A is non-empty, each A_i is non-empty. Does the converse hold?
(d) What is the relation between the set $A \cup B$ and the Cartesian product of the sets $A_i \cup B_i$? What is the relation between the set $A \cap B$ and the Cartesian product of the sets $A_i \cap B_i$?

4. Let $n, m \in \mathbb{Z}_+$. Let $X \neq \emptyset$.

(a) If $m \leq n$, find and injective map $f : X^m \to X^m$.
(b) Find a bijective map $g : X^m \times X^n \to X^{m+n}$.
(c) Find an injective map $h : X^n \to X^\omega$.
(d) Find a bijective map $k : X^n \times X^\omega \to X^\omega$.
(e) Find a bijective map $l : X^\omega \times X^\omega \to X^\omega$.
(f) If $A \subset B$, find and injective map $m : (A^\omega)^n \to B^\omega$.

5. Which of the following subsets of $\mathbb{R}^\omega$ can be expressed as the Cartesian product of subsets of $\mathbb{R}$?

(a) $\{\mathbf{x} \mid x_i \text{ is an integer for all } i\}$.
(b) $\{\mathbf{x} \mid x_i \geq i \text{ for all } i\}$.
(c) $\{\mathbf{x} \mid x_i \text{ is an integer for all } i \geq 100\}$.
(d) $\{\mathbf{x} \mid x_2 = x_3\}$.